

Effects of an arbuscular mycorrhizal fungus on *Amorpha fruticosa* roots and soil preferential flow in an arid area of opencast coal mine waste

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ABSTRACT

Preferential flow plays a key role in soil hydrological processes in arid and semi-arid areas of opencast coal mine waste. Inoculation with arbuscular mycorrhizal (AM) fungi can significantly affect vegetation root growth and improve soil physical structure. However, the impact on preferential flow in new areas of waste remains poorly understood. Here, the effects of AM fungal inoculation on root spatial distribution and preferential flow within the waste area at Heidaigou opencast coal mine in China were analyzed where *Amorpha fruticosa* was grown for revegetation. A staining tracer method and a grid sampling method were used. Preferential flow pathway development in the newly formed waste areas was affected mainly by *A. fruticosa* root systems, with closer proximity to *A. fruticosa* resulting in more preferential flow pathways. Inoculation with the AM fungus *Funneliformis mosseae* significantly increased preferential flow development. In the profile closest to the *A. fruticosa*, the dye coverage increased by 59 %, the uniform infiltration depth by 73 %, and the maximum stained depth by 80 %. Plant roots occurred mainly at 0–20 cm soil depth, accounting for ~ 80 % of the total root length of the entire root system. In the profile closest to *A. fruticosa*, inoculation with *F. mosseae* increased the total number of roots and the root length density by 32 %. The preferential flow pathways were developed only in roots with diameters of > 2 mm and more widely distributed at both 0–10 and 10–20 cm soil depths. Substrate infiltration was influenced mainly by roots with diameters of < 4 mm but deep infiltration was more dependent on roots with diameters > 2 mm. Mycorrhizal hyphae contributed to the preferential flow. Inoculation with the AM fungus increased the preferential flow of the newly formed drainage sites and this effect should be considered in the ecological restoration of opencast coal mine wastes in arid and semi-arid areas.

1. Introduction

Surface mining has had adverse environmental impacts leading to the accumulation of coal gangue which, in turn, has resulted in the formation of disposal sites that disrupt ecosystems and soil structure and render surface soils infertile (Neugirg et al., 2016). It is worth noting that opencast coal mine dumps cover 30–50 % of the total land of mines. China has surface coal mines primarily located in its ecologically fragile northwest region characterized by arid climates, limited water resources, high evaporation rates, and significant impacts of soil moisture on plant growth (Yang et al., 2022). In addition, the mechanical compaction resulting from large-scale machinery operations in opencast coal mine dumps leads to changes in soil structure, hampering plant root

growth, reducing water and nutrient uptake, limiting aeration, and ultimately diminishing plant yields (Abu-Hamdeh, 2003; Kim et al., 2010). Water is a critical factor in land reclamation and ecological restoration, but arid and semi-arid regions suffer from severe water scarcity, with total water resources accounting for only 3.9 % of the national total. Compacted soil in these areas does not effectively retain rainfall, often leading to water loss through runoff. Moreover, high temperatures following rainfall events and rapid soil water evaporation make the conditions challenging for plant growth. This accelerated evaporation further depletes water moisture content (Wang et al., 2021b). Consequently, water is a major limiting factor in vegetation recovery in these areas, making efficient use of precipitation a high priority.

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Preferential flow is the uneven and erratic movement of water flow that bypasses most of the porous matrix network of soils (Bouma and De Laet, 1981) and is important in soil hydrology (Allaire et al., 2009). Despite only accounting for 0.23–20 % of the total soil volume (Alaoui and Helbling, 2006), preferential flow pathways exhibit high water transport capacity, with macropores capable of transporting 74–100 % of soil moisture. Preferential flow is important in shaping runoff, infiltration, partitioning, and nutrient transport (Beven and Germann, 2013). It is influenced at the pore scale by various factors including soil texture and structure (Reading et al., 2012; Sanders et al., 2012; Shein, 2010), plant root systems (Bargués Tobella et al., 2014; Qiu et al., 2023; Zhang et al., 2015a), gravels (Hou et al., 2023; Zhang et al., 2016a), animals, and microorganisms (Han et al., 2015; Yagüe et al., 2016). Channels around roots are vital to soil macropores and are crucial preferential flow pathways (Jáčka et al., 2021). The living plant root system establishes a root-soil interface critical for water uptake and nutrient acquisition during root growth. Simultaneously, the pore spaces between the root-soil interface serve as continuous preferential flow pathways (Bargués Tobella et al., 2014; Ceccon et al., 2011; Tracy et al., 2011). Large pores resulting from the microbial decomposition of root systems also become continuous preferential flow pathways (Backnäs et al., 2012; Zhang et al., 2015a; Zhang et al., 2016b). Plant roots are important in improving soil pore structure, increasing soil fertility, enhancing soil activity, and altering preferential flow pathways (Bochet, 2015; Singh et al., 2015). Numerous studies have explored preferential flow in agricultural soils due to its potential for water and fertilizer loss and the contamination of groundwater by substances such as pesticides (Perkins et al., 2011; Steenhuis et al., 1988; Yi et al., 2020). In arid and semi-arid opencast coal mine waste areas, promoting this type of infiltration to allow precipitation to bypass the soil matrix and rapidly enter the soil profile is important in maintaining soil moisture levels essential for plant growth. Investigating the relationship between root distribution in the soil and soil preferential flow is critical. Understanding how root systems affect soil preferential flow under compacted soil conditions is also essential.

Potentially symbiotic relationships between root systems and arbuscular mycorrhizal (AM) fungi are widespread and the elongated, highly branched hyphae can regulate root morphology and development, enhancing plant nutrient uptake (He and Zhong, 2012; Shen et al., 2020). Mycorrhizal mycelium competes with the plant for limited resources, creating an extensive underground network for nutrients and water transfer between the soil and plant (Chen et al., 2018; Shi et al., 2011). AM fungi also secrete glomalin-related soil protein (GRSP) which acts as a binding agent in the soil (Bedini et al., 2009; Rillig, 2004). The entanglement of AM fungal mycelium and the cementing action of GRSP contribute to the formation and stabilization of soil aggregates (Rillig and Mummey, 2006). In arid and semi-arid regions the natural ecological recovery of opencast coal mine wastes is very slow. Reclamation and revegetation are effective approaches, supported by field and laboratory studies demonstrating that the inoculation of reclaimed vegetation with AM fungi enhances soil nutrient status and promotes plant growth (Nunez-Mir et al., 2015; Li et al., 2015). Consequently, the application of AM fungal inoculants is an effective strategy for restoring degraded ecosystems such as deserts and drainage areas (Orlowska et al., 2010; Taheri and Bever, 2011). Although the influence of AM fungi inoculation on morphological characteristics of root systems and soil aggregates can dramatically alter soil pore structure, their impact on the generation of preferential flow pathways in arid and semi-arid regions remains poorly investigated.

Here, the role of AM mycorrhizal associations in the hydrological processes of opencast coal mine waste areas was investigated by examining the spatial distribution of plant root systems and preferential flow. The aim was to determine how AM fungal inoculation of plants in compacted soil conditions affects soil preferential flow, the morphology and spatial distribution of root systems, and the range of root diameters involved in preferential flow. The results will contribute to addressing

the low utilization rate of natural precipitation in open pit dumps in arid and semi-arid regions and provide a theoretical basis for applying microorganisms in ecological restoration.

2. Materials and methods

2.1. Study area

The study area is at the Heidaigou opencast coal mine dump (111°13′–111°20′E, 39°43′–39°49′N) in Xuejiawan town, Jungar banner, Ordos city, Inner Mongolia Autonomous Region, northwest China. It has a semi-arid continental climate in the middle temperate zone with a mean annual rainfall of ~ 400 mm and a mean annual potential evapotranspiration of ~ 2000 mm. Precipitation occurs primarily between July and September when it accounts for 60–70 % of the annual precipitation. The area maintains an average annual temperature of 7.2 °C, a maximum permafrost depth of 1490 mm, and an annual sunshine duration of 3119 h. The above data comes from China Meteorological Data Network (<http://data.cma.cn/>). Positioned in the northeastern sector of the Ordos Plateau on the Loess Plateau, this region is characterized by hilly terrain covered with loess soils. The soil at the dump site is a newly formed, weakly alkaline (pH=7.5), with low organic matter content (~7 g/kg, available phosphorus (5 mg/kg), and available potassium (110 mg/kg). The soil is nutrient-poor, highly compacted due to mechanical operation and exhibits slow infiltration. The native vegetation consists of *Artemisia gmelinii*, *Artemisia giraldii*, *Stipa bungeana*, and *Bothriochloa ischaemum* at different stages of ecological succession.

2.2. Plot selection, sample collection, and tracer experiments

The experimental plots were located within the inner dump of Heidaigou coal mine. Established in April 2018, the area was planted with *A. fruticosa* seedlings, with an average seedling height of 50 cm. Two test plots were designated: an AM plot with AM fungal inoculation and a control (CK) plot without inoculation. Each plot measured 44 m in length and 34 m in width, with an 80 m gap separating them (Fig. 1). A total of 16 rows of seedlings were planted in each plot, each row containing 21 plants. The inoculant was *Funneliformis mosseae* and was applied directly to the plant roots at 50 g per plant. After planting, each plot was irrigated to reach the soil maximum saturated water-holding capacity once a week for one month.

In August 2022 the growth of *A. fruticosa* in each plot was assessed and the understory herbaceous community succession was investigated. Three representative sample sites were selected, each centered on an *A. fruticosa* plant with similar understory vegetation indicators and comparable particle size distribution and related soil physical properties across different treatments (Table 1). No precipitation had occurred in the test area for at least two weeks before sampling and conducting the staining tracer test. The field tracer test was conducted continually from August 25 to September 10, 2022.

To prepare for the tracer test the surface vegetation within a 1 × 1 m area was removed and the surface was leveled. A metal frame 100 cm in length, 100 cm in width, and 40 cm in height was inserted into the soil to a depth of 20 cm and the gaps at the edges were carefully filled and tamped. Brilliant blue FCF was added to water at a concentration of 4.0 g L⁻¹. A total of 60 L of the solution was evenly sprayed into the metal frame using a sprinkler head within 1 h to simulate rainfall with an intensity of 60 mm h⁻¹ (the average annual maximum rainfall intensity in the study area). The solution was uniformly applied to a 1 × 1 m area centered on the experimental plants to avoid border effects (Fig. 2a). The metal frame was covered with plastic film to prevent evaporation. After 24 h a vertical profile was excavated. To eliminate boundary effects, 5 cm was removed from both sides, resulting in a 90-cm-wide and 40-cm-deep profile. The vertical profiles were placed at distances of 5 cm (V3), 25 cm (V2), and 45 cm (V1) from the plants (Fig. 2b). Each profile was carefully trimmed with a trowel for better

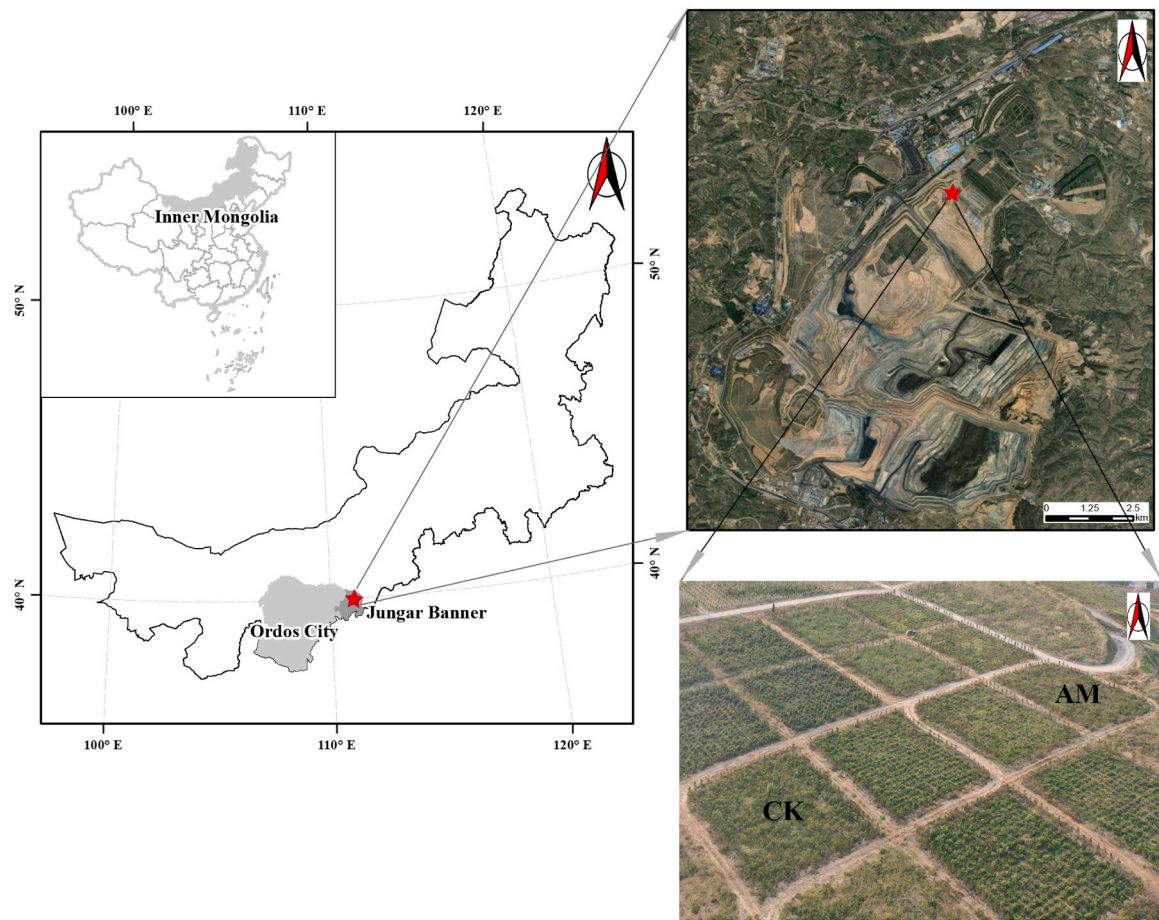


Fig. 1. Study site location AM: *Amorpha fruticosa* inoculated with the AM fungus *Funneliformis mosseae*; CK: *Amorpha fruticosa* without AM fungal inoculation (Control).

Table 1
Basic information on the sampling plots.

Sample	Plant height (cm)	Canopy (cm)	Abundance	Bulk density (g cm ⁻³)	Initial soil water content (%)	Particle-size distribution (%)		
						Clay (<0.002 mm)	Silt (0.002–0.05 mm)	Sand (0.05–2 mm)
AM1	170	220 × 185	29	1.59	6.75	5.76	53.38	40.86
AM2	175	160 × 150	38	1.61	7.51	6.48	50.06	43.46
AM3	169	260 × 160	30	1.57	6.55	6.14	51.72	42.14
CK1	130	170 × 150	35	1.55	6.73	6.18	49.48	44.33
CK2	110	180 × 135	32	1.60	6.02	6.68	50.62	42.70
CK3	130	153 × 100	39	1.54	6.47	6.50	53.47	40.02

AM: *Amorpha fruticosa* inoculated with the AM fungus *Funneliformis mosseae*; CK: *Amorpha fruticosa* without AM fungal inoculation (Control); Particle size distribution of the soil was categorized using the system of the United States Department of Agriculture (USDA).

visualization of the stained areas, and the calibration frame was placed at the soil profile boundary for geometric correction at later stages. Digital photographs of the soil profiles were taken for preferential flow path (PFP) analysis.

Root samples were collected from the vertical soil profiles (10 cm × 10 cm) using the grid method. A root sample collector with a volume of 100 cm³ was used to take root samples in each grid. The roots were then thoroughly washed, passed through a 2-mm sieve, and scanned using a ScanMaker i800 plus root scanner (Microtek, Shanghai, China). Soil samples were taken from each grid and mycelial density in the soil was determined using the grid-intersect method (Abbott et al., 1984).

2.3. Image analysis and preferential flow parameters

The field-acquired staining profile images underwent several processing steps. First, the images were spatially corrected using ArcGIS 9.3 (Esri, Redlands, CA) according to the correction box, followed by retrograde cropping and color adjustment using Photoshop CC2019 (Adobe, San Jose, CA). The processed images were then transformed into a numerical matrix consisting of 0 and 255 with one pixel per mm, where 255 indicates stained areas and zero denotes non-stained areas. Subsequently, various preferential flow parameters were calculated using the resulting data as described previously (Bargués Tobella et al., 2014; Qiu et al., 2023; Van Schaik, 2009):

Dye coverage (DC, %) was defined as the percentage of the dye-stained area relative to the sum of the dye-stained and non-stained

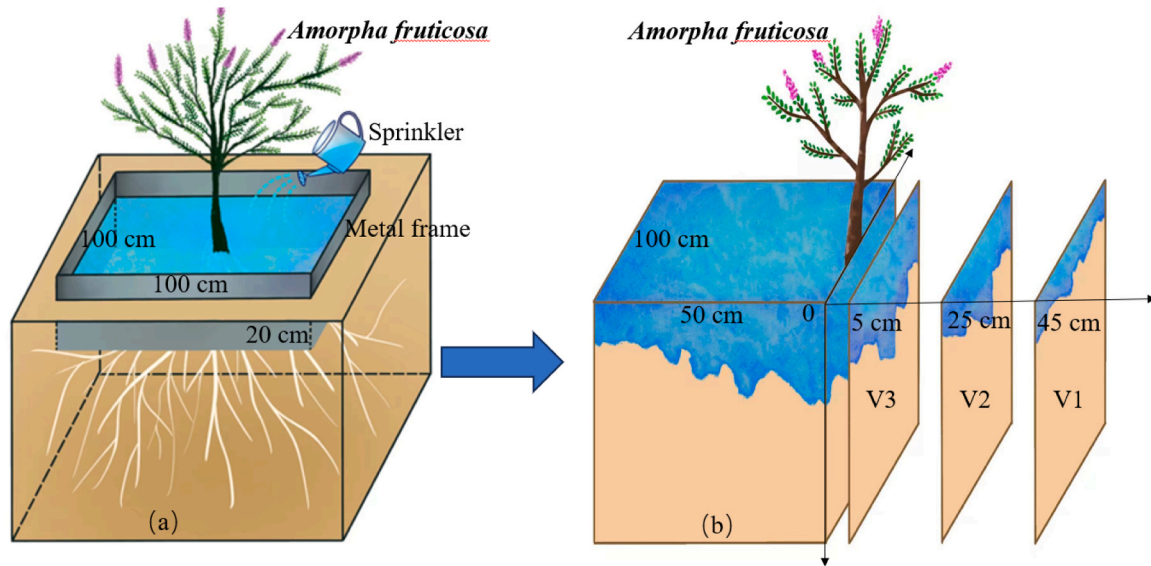


Fig. 2. The position of the metal frame and the spraying of the brilliant blue FCF solution are shown in (a), and (b) schematic drawings of three vertical soil profiles at the same sample site are shown.

areas. A higher DC indicates a lower degree of preferential flow.

$$DC = 100 \times \left(\frac{D}{D + ND} \right) \quad (1)$$

where D and ND are the dye-stained area (cm^2) and the non-stained area (cm^2), respectively. Coefficient variation (CV) was defined as the difference between soil-stained and non-stained areas due to preferential flow in the vertical soil profile.

$$CV = \frac{\sqrt{\frac{1}{n-1} \sum_{i=1}^n (DC_i - \overline{DC})^2}}{\frac{1}{n-1} \sum_{i=1}^n DC_i} \quad (2)$$

where n is the number of soil layers in the vertical profile, DC_i is the stained area ratio (%) of the soil in layer i^{th} , and $\overline{DC}$ is the mean value of the stained area ratio (%). Smaller CV values suggest a higher degree of preferential flow development.

Uniform infiltration depth (UniFr, cm) was defined as the depth at which the dye coverage falls below 80 %. Soils with a lower UniFr indicate a higher degree of preferential flow. UniFr was obtained by counting stained pixels.

Maximum stained depth (MSD, cm) was the maximum depth to which water infiltrates through the preferential flow pathways and was calculated by counting stained pixels.

Preferential flow fraction (PF-fr, %) was defined as the fraction of the total infiltration that flows through the preferential flow pathways:

$$PF - fr = 100 \times \left(1 - \frac{UniFr \times 30}{TotStAr} \right) \quad (3)$$

where UniFr is the uniform infiltration depth (cm) and TotStAr is the total stained area (cm^2).

Length index (LI) was defined as the sum of the absolute differences between consecutive dye coverage values at different depths in a soil profile. Soils with higher LI values exhibit a higher degree of preferential flow:

$$LI = \sum_{i=1}^n |DC_{i+1} - DC_i| \quad (4)$$

where i represents a given layer at a stained profile, and n is the total

number of layers. The vertical profile was divided at 20-mm intervals.

2.4. Statistical analysis

Statistical analysis was conducted using the SPSS Statistics 26 software package (IBM, Armonk, NY). The effects of the AM fungus on preferential flow parameters and root lengths at each radial level were assessed using one-way analysis of variance (ANOVA). The relationships between preferential flow parameters and root lengths at each diameter level were analyzed using Pearson correlation analysis. Graphs and data fitting were generated using Origin 2021 software. Profile root length distribution maps were drawn using GraphPad Prism 9 software.

3. Results

3.1. Characteristics of preferential flow

Fig. 3 shows the staining images of different profiles showing significant differences between treatments and different sampling sites in the same treatment. First, preferential flow in the mycorrhizal and control plots was predominantly characterized by large pore flow. Across the six samples in the two treatments, proximity to the plants resulted in more complex staining patterns and a higher degree of preferential flow development. In profiles situated 45 cm from the plants, staining areas were small, shallow in depth, and distant from *A. fruticosa*. In addition, these areas were covered by herbaceous plants. These findings indicate that the root systems of herbaceous plants do not generate enough large pores in the newly formed opencast coal mine waste areas and cannot establish preferential flow pathways. In profiles located 5 cm from the plants the stained areas were concentrated in the middle of the profile, and this is attributable to the presence of *A. fruticosa* in the middle position. These findings indicate that *A. fruticosa* plays an important role in shaping preferential flow pathways, with its root systems being the primary contributor to their formation.

Fig. 4 shows the changes in staining area ratio with soil depth. In the same treatment, three replicates exhibited smaller differences in the V1 profile but larger differences in the V2 and V3 profiles. These differences were more obvious in the AM treatment, indicating the substantial influence of the root system of *A. fruticosa* on the distribution of preferential flow across various soil depths. The V1 profile, farthest from

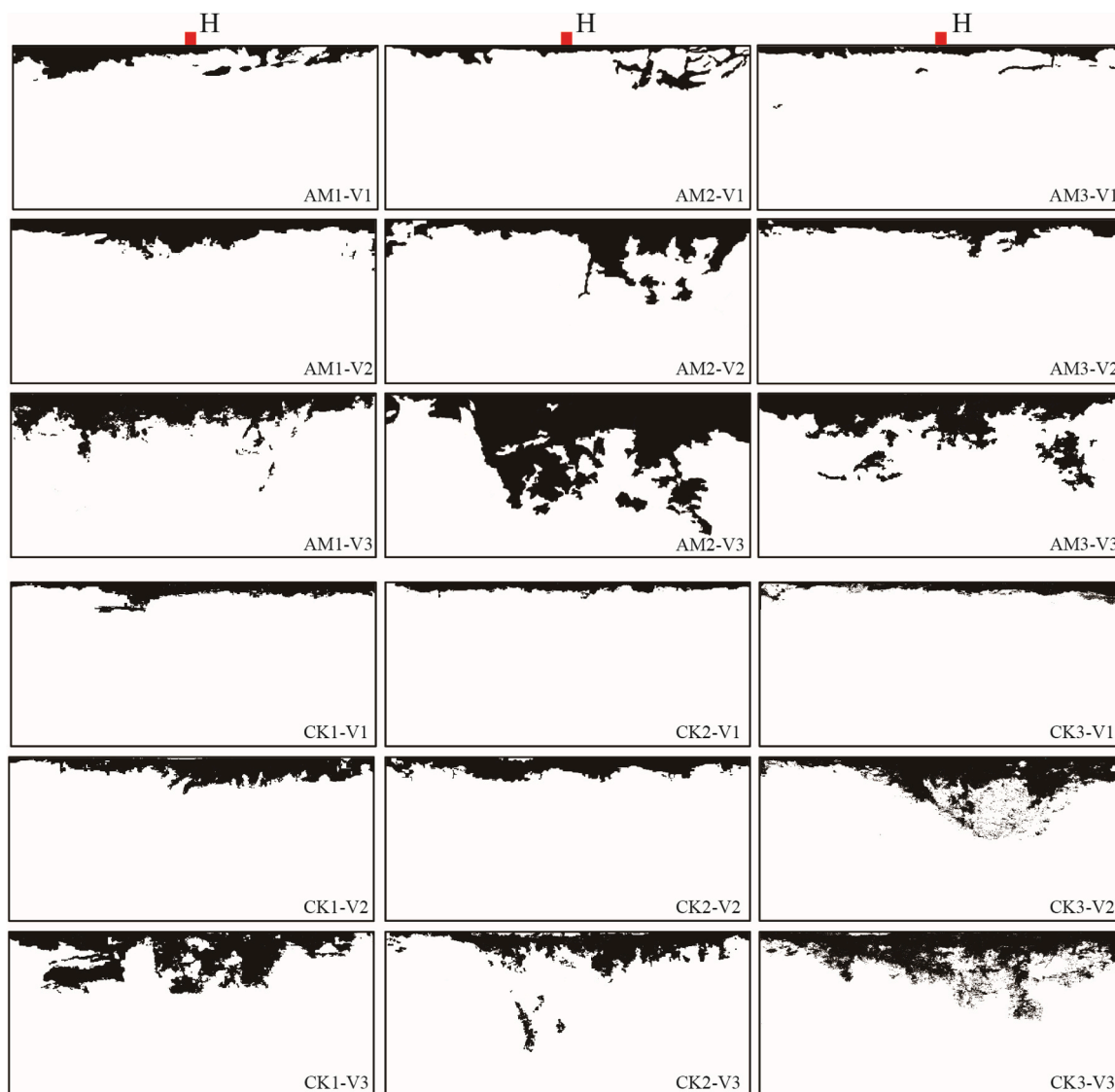


Fig. 3. Sample staining patterns at the six experimental sites. AM1, AM2, and AM3 are the three *Amorpha fruticosa* inoculation plots; CK1, CK2, and CK3 are the three *Amorpha fruticosa* control plots; V1, V2, and V3 are the vertical profiles at distances of 45 cm, 25 cm, and 5 cm from *Amorpha fruticosa*, respectively; and H is the location of *Amorpha fruticosa* in the profile.

A. fruticosa, shared a similar pattern of changes in staining area ratio with increasing soil depth between the AM treatment and the control due to the presence of only understory herbaceous vegetation in this area. The resemblance of the staining area ratio in the V1 profile to that of the control suggests that the root system of herbaceous plants has a limited effect on improving the deep soil structure of the newly developed dump, highlighting the limitations of herbaceous plants in enhancing soil structure in these areas.

Fig. 5 demonstrates the significance of differences in the six preferential flow parameter indexes. In the inoculated and control plots, DC (Fig. 5a) increased as proximity to the plant increased. Moreover, the DC of the AM treatment was higher than that of the control in all profiles. Notably, in the V3 profile the DC of the treatment (24.06) was 1.59 times that (15.05 %) of the control ($P < 0.05$), showing a significant difference. Moreover, the CV of the stained area ratio was also the smallest in the V3 profile (Fig. 5b), indicating greater preferential flow development near *A. fruticosa*. UniFr was smaller in V1 and V2 profiles of both AM and CK treatments (Fig. 5c), ranging between 1.70 and 3.27 cm. In the V3 profile, the UniFr of the AM treatment was 6.07 cm, 1.73 times that (3.50 cm) of the control ($P < 0.05$). The MSD was 15–80 % higher in the AM treatment than in the CK treatment and reached a significant

difference ($P < 0.05$) in the V3 profile (Fig. 5d). Pf-fr was significantly different solely between the inoculated and control plots in the V1 profile (Fig. 5e). LI was significantly higher in both V2 and V3 profiles of the AM plots compared to the controls ($P < 0.05$), with the highest increase of 16 % (Fig. 5f). In summary, the preferential flow in the newly opencast coal mine dump was affected mainly by *A. fruticosa*, with limited impact from understory herbaceous plants. The highest degree of preferential flow development was observed in the profile closest to *A. fruticosa*. The AM treatment significantly increased the degree of preferential flow development, particularly in the UniFr and MSD which increased by 73–80 % in the AM treatment.

3.2. Effect of AM fungal inoculation on the spatial distribution of root length density (RLD)

The spatial distribution of RLD of the root system differed greatly in different profiles (Fig. 6). The closer the profile to *A. fruticosa* the wider the distribution of the plant root system. In the AM treatment the average RLD of the V3 profile was $5052 \text{ mm (100 cm)}^{-3}$, 3.51 and 1.62 times that of $1441 \text{ mm (100 cm)}^{-3}$ of the V1 profile and $3128 \text{ mm (100 cm)}^{-3}$ of the V2 profile, respectively. In the CK treatment the mean

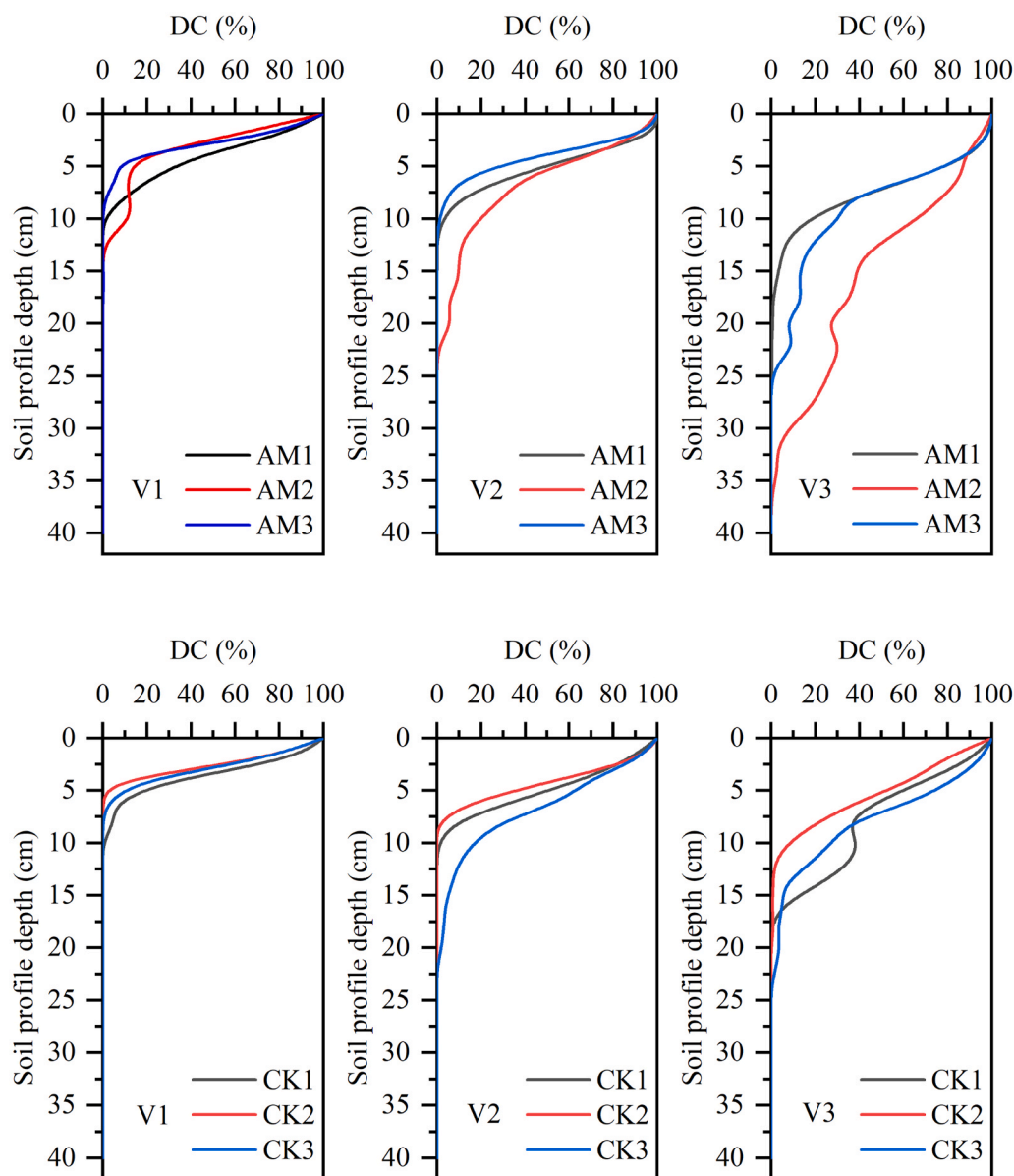


Fig. 4. Differences in dye coverage with soil depth. AM1, AM2, and AM3 are the three *Amorpha fruticosa* inoculation plots; CK1, CK 2, and CK3 are the three *Amorpha fruticosa* control plots; V1, V2, and V3 are the vertical profiles at distances of 45 cm, 25 cm, and 5 cm from *Amorpha fruticosa*, respectively. DC, dye coverage.

RLD of the V3 profile of CK was $3832 \text{ mm (100 cm)}^{-3}$, 1.69 and 1.12 times that of $2271 \text{ mm (100 cm)}^{-3}$ of the V1 profile and $3422 \text{ mm (100 cm)}^{-3}$ of the V2 profile respectively. The AM treatment reduced the mean RLD of V1 and V2 profiles to only 59 and 92 % of that of the control. However, the AM treatment increased the RLD of the V3 profile by 32 %, indicating that AM promoted the root system of *A. fruticosa* itself. The root system in profiles farther from *A. fruticosa* consisted mainly of understory herbaceous roots, and inoculation with the AM fungus may have inhibited the growth of these understory herbaceous roots. In both AM and CK treatments the root system was mainly distributed in the 0–20 cm soil depth, accounting for ~80 % of the total root system.

Fig. 7 provides a detailed analysis of the root systems in PFPs and soil matrix (SM) zones. The root systems were divided into eight diameter classes and analyzed based on total RLD at different diameter classes in various soil depths. Fine roots with a diameter ranging from 0 to 1 mm were widely distributed in both stained and unstained zones, with root length density consistently exceeding $100 \text{ mm (100 cm)}^{-3}$. When the root diameter exceeded 2 mm, differences between PFPs and SM became

more apparent. Three scenarios were observed in terms of root diameter composition in the unstained zone compared to the stained zone. These scenarios were: 1) roots > 2 mm in diameter were barely distributed in the 0–10 cm and 10–20 cm layers, with root length density < $10 \text{ mm (100 cm)}^{-3}$; 2) roots of > 2 mm in diameter were distributed in the 0–10 cm layer but less in the 10–20 cm layer; and 3) roots of > 2 mm in diameter were barely distributed in the 0–10 cm layer, and were actually more abundant in the 10–20 cm layer and, they were unable to form a preferential flow pathway. Therefore, the preferential flow pathways were established only when roots with diameter > 2 mm were more distributed in both the 0–10 and 10–20 cm soil depths.

Fig. 8 demonstrates that the RLD of root systems in different diameter classes in the PFPs was significantly higher than that in the SM zone ($P < 0.05$). In particular, root systems > 7 mm were absent from the SM zone. The highest RLD in the PFPs was about $415 \text{ mm (100 cm)}^{-3}$, with the RLD of 0–7 mm root systems in the PFPs being 4–11 times that in the SM zone. Compared to that in PFPs of the control the RLD of roots of all size classes in the PFPs of the AM treatment increased by 34–196 %, with the highest increase (196 %) seen in roots 2–3 mm in diameter.

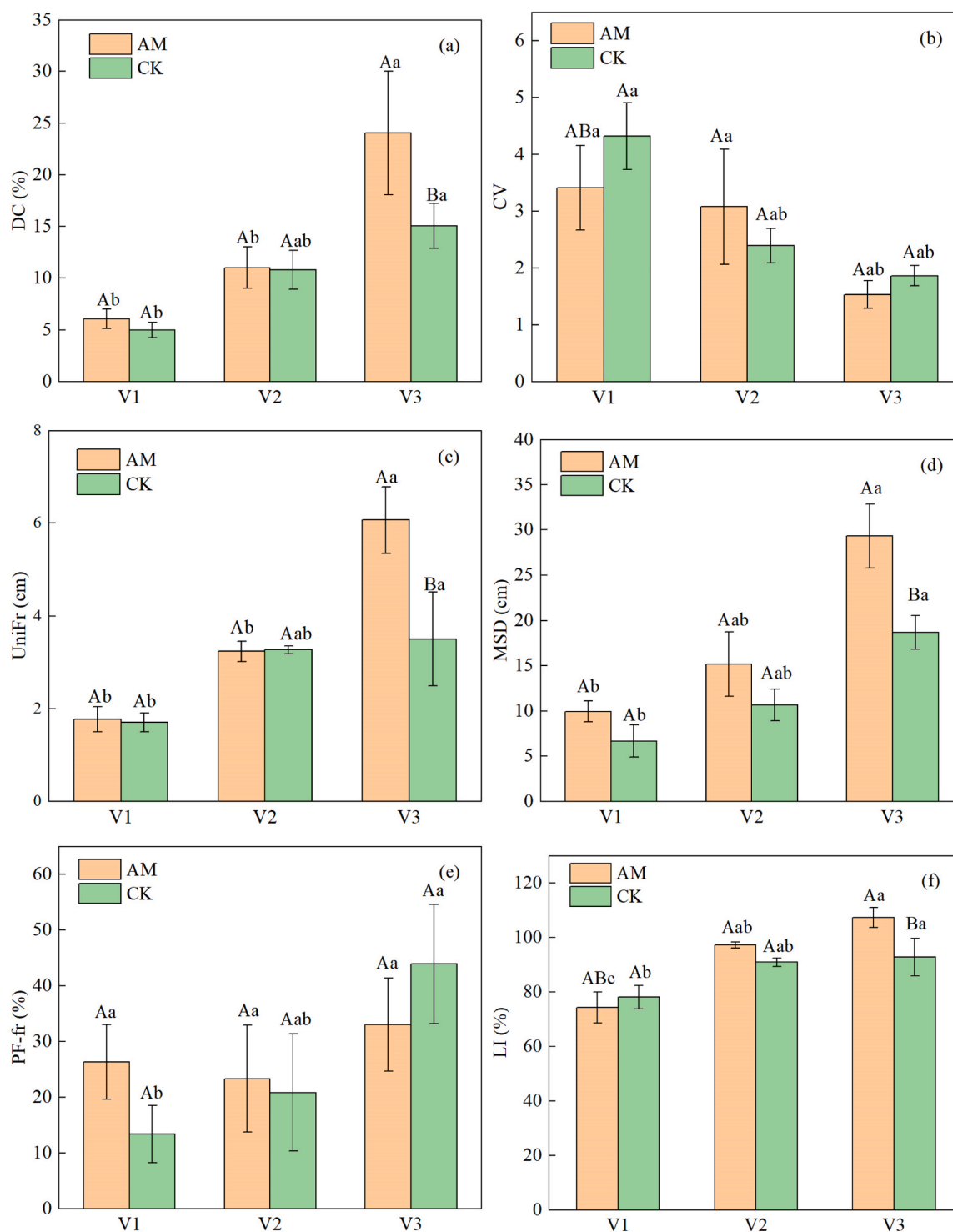


Fig. 5. The preferential flow parameters for different treatments and profiles. (a) coefficient of variation (CV), (b) dye coverage (DC), (c) uniform infiltration depth (UniFr), (d) maximum stained depth (MSD), (e) preferential flow fraction (PF-fr), and (f) length index (LI). Different lowercase letters indicate significant differences at $P < 0.05$ between different profiles of the same treatment, and different capital letters indicate significant differences at $P < 0.05$ between different treatments of the same profile.

3.3. Effect of RLD on preferential flow indexes

RLD of all root classes had a significant effect on the preferential flow index (Table 2). The DC shows a significant positive correlation with RLD in most root classes, while the CV shows a significant negative correlation with RLD in all root classes, indicating that plant roots of all classes played a facilitating role in soil infiltration and preferential flow

pathway development. The UniFr was mainly affected by root systems with diameters < 4 mm. The MSD was significantly and positively correlated only with RLD of root systems with diameters of 2–3 mm. The LI was significantly and positively correlated only with the RLD of root systems with diameters > 7 mm. Moreover, $RLD_{2-3\text{ mm}}$ was significantly and negatively correlated with the CV of the staining area ratio ($P < 0.05$) (Fig. 9a) which explained 64.4 % of the variation.

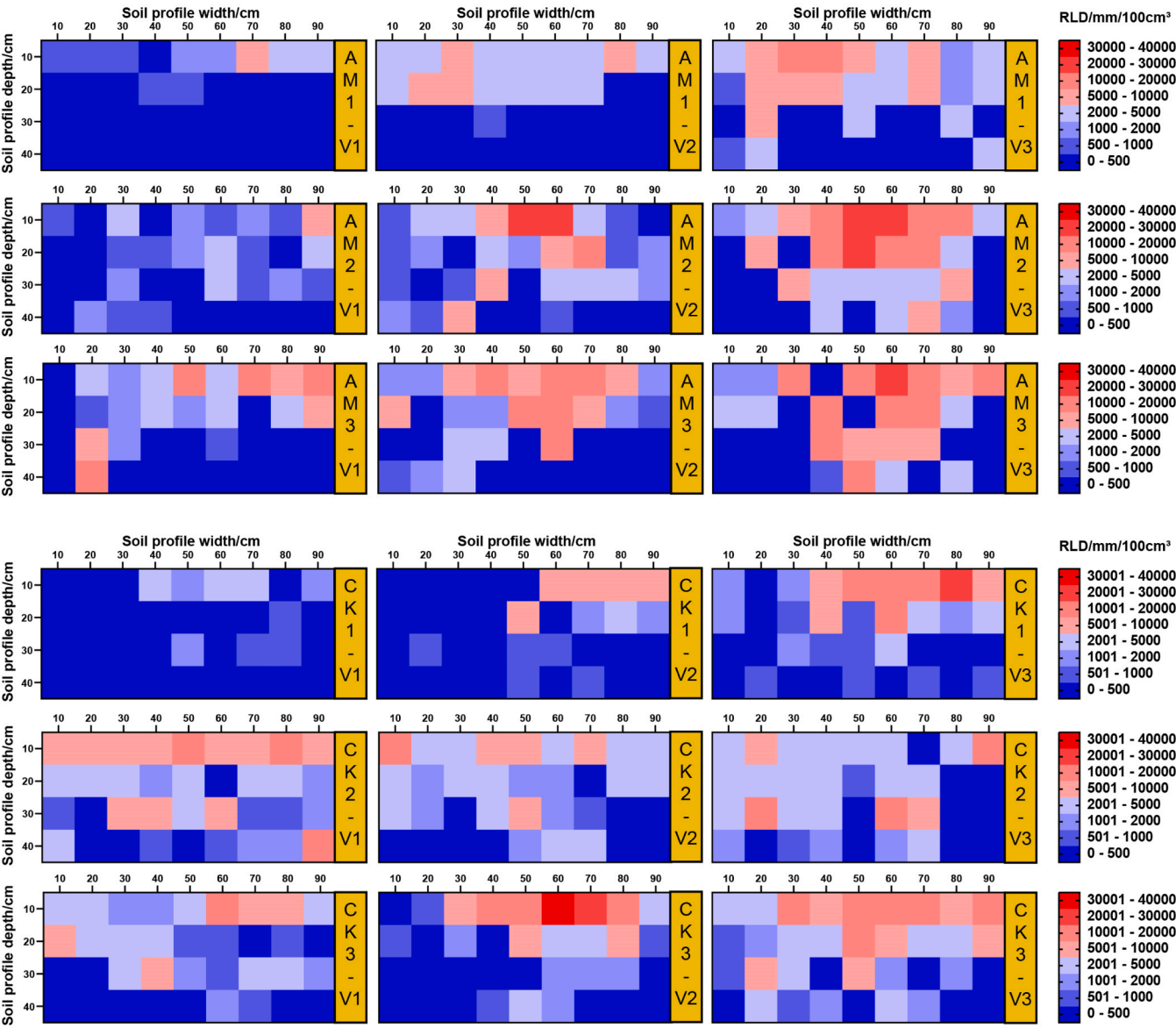


Fig. 6. Distribution of root length density in soil profiles at different distances of different treatments. AM1, AM2, and AM3 are the three *Amorpha fruticosa* inoculation plots; CK1, CK2, and CK3 are the three *Amorpha fruticosa* control plots; V1, V2, and V3 are the vertical profiles at distances of 45 cm, 25 cm, and 5 cm from *Amorpha fruticosa*, respectively.

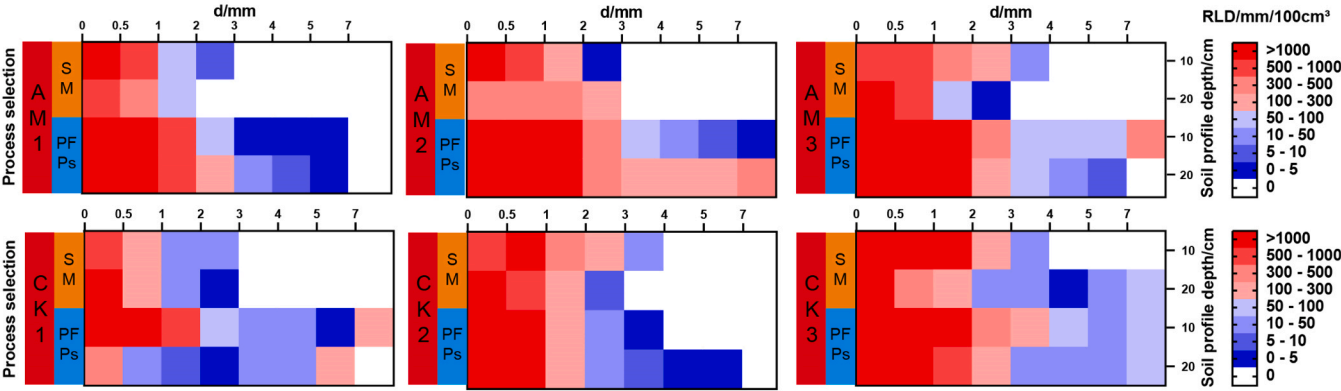


Fig. 7. Composition of root length distribution at different root size classes. SM is the soil matrix, and PFPs are the preferential flow pathways.

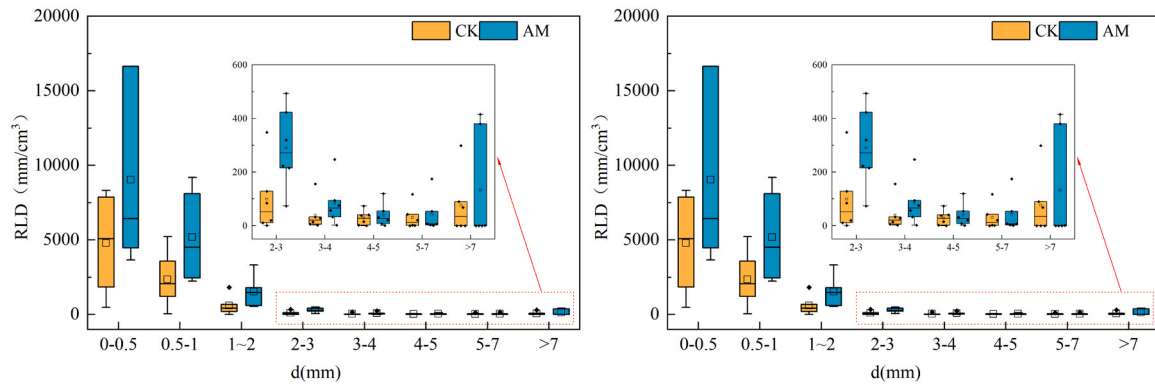


Fig. 8. Root length distribution characteristics of different root size classes. SM is the soil matrix, and PFPs are the preferential flow pathways. RLD, root length density. d, diameter.

Table 2

Correlations between the root length density of different diameter classes and the preferred flow parameters.

Parameter	RLD 0–0.5mm	RLD 0.5–1mm	RLD 1–2mm	RLD 2–3mm	RLD 3–4mm	RLD 4–5mm	RLD 5–7mm	RLD >7 mm
CV	–.923**	–.875*	–.876*	–.846*	–.886*	–.843*	–.936**	–.830*
DC	.972**	.931**	.919**	.901*	.916*	0.785	.854*	0.678
UniFr	.885*	.881*	.861*	.900*	.883*	0.785	0.684	0.473
MSD	0.773	0.811	0.788	.842*	0.727	0.485	0.366	0.196
PF-fr	–0.121	–0.186	–0.173	–0.291	–0.215	–0.264	0.049	0.135
LI	0.589	0.637	0.677	0.716	0.598	0.655	0.619	.816*

CV, coefficient variation; DC, dye coverage; UniFr, uniform infiltration depth; MSD, maximum stained depth; PF-fr, preferential flow fraction; LI, length index. RLD_{0–0.5mm}–RLD_{>7 mm} represents the length density of roots with different diameters. * $P < 0.05$; ** $P < 0.01$.

RLD_{2–3 mm} was significantly and positively correlated with DC, UniFr, and MSD ($P < 0.05$) (Fig. 9b–d), explaining 76.5, 76.2, and 63.5 % of the variance, respectively. RLD_{>7 mm} was significantly positively correlated with LI ($P < 0.05$) (Fig. 9e), explaining 58.2 % of the variance. In summary, preferential flow pathway development in the four-year reclaimed opencast coal mine waste area was strongly influenced by the root systems, with the fine roots (< 2 mm) playing an important role in the surface soil and affecting the substrate infiltration with the coarse roots (> 2 mm) playing an important role in soil infiltration in the deeper layers (20–40 cm depth).

4. Discussion

4.1. Preferential flow characteristics of the opencast coal mine waste

The dye tracer test is a widely used method to study preferential flow, where the dye's color density on the soil profile represents the flow paths, and the dye concentration in different profile areas indicates solute movement (Mei et al., 2018; Yi et al., 2020). In this study, a concentration of $4 \text{ g} \cdot \text{L}^{-1}$ of brilliant blue FCF dye was selected to ensure visibility even after dilution and adsorption, and the preferential flow paths were identified by analyzing the stained areas. The study reveals differences in stained areas and maximum stained depth between different treatments and among different profiles of the same treatment. These findings align with a previous report (Jarvis, 2020) in which the initial soil water content affected preferential flow depth and that higher soil water contents were associated with less lateral water infiltration and increased macropore flow into deeper soil layers. Here, the average soil bulk density of the sampling sites was $1.60 \text{ g} \cdot \text{cm}^{-3}$ (Table 1), much higher than the 1.00 – $1.30 \text{ g} \cdot \text{cm}^{-3}$ in the natural state of the region (Wang et al., 2021a). Compaction hinders soil substrate infiltration and lateral infiltration, allowing more water to flow into the macropores. The stained area and depth were notably constrained in the V1 and V2 profiles distant from *A. fruticosa* (Fig. 3). Furthermore, the stained area ratio versus depth change curves in the V1 profile consistently

demonstrate analogous patterns across all six sample points (Fig. 4), implying minimal water filtration in these areas. Even under conditions of reduced water infiltration, preferential flow pathways remained intact, suggesting the substantial impact of compaction in the opencast coal mine waste area on substrate infiltration. These findings align with previous research showing that although compaction reduces micropores, macropores resistant to vertical compression predominantly facilitate infiltration (Alaoui and Helbling, 2006).

Herbaceous root systems may also produce new preferential flow pathways post-revegetation, although grasslands have the least soil infiltration capacity relative to arboreal and shrubby woodlands (Qiu et al., 2023). Here, the profile closest to *A. fruticosa* (V3) exhibited the highest preferential flow development, with all indicators performing better than in the other profiles.

4.2. Effects of AM fungus inoculation on plant root distribution

Inoculation with the AM fungus significantly increased the number of roots in the soil. This effect was more pronounced in proximity to *A. fruticosa*, aligning with findings from other studies (Bi et al., 2019; Cosme and Wurst, 2013; Hashem et al., 2019). The plant root systems decreased with increasing soil depth, a trend consistent with the findings of DuPont et al. (2014) and Ferchaud et al. (2015). Furthermore, the root systems were concentrated mainly at 0–20 cm soil depth, accounting for > 80 % of the total root system and in agreement with previous observations (Ni et al., 2015; Raizada et al., 2013; Zhang et al., 2017). The abundance of roots in the upper soil improved soil structure, augmented soil nutrient exchange, elevated soil activity, increased soil porosity, and contributed to both substrate infiltration and preferential flow (Zhang et al., 2017). Inoculation with AM fungi significantly increases mycelial density (HD) in soils (Bhantana et al., 2021; Gujre et al., 2021). In the AM-treated areas, HD around *A. fruticosa* increased by 24.6–55.0 % (Fig. 10). This extensive mycelial network facilitated plant water and nutrient uptake, contributing to rapid plant growth (Chen et al., 2018). Moreover, the plants inoculated with the AM fungus

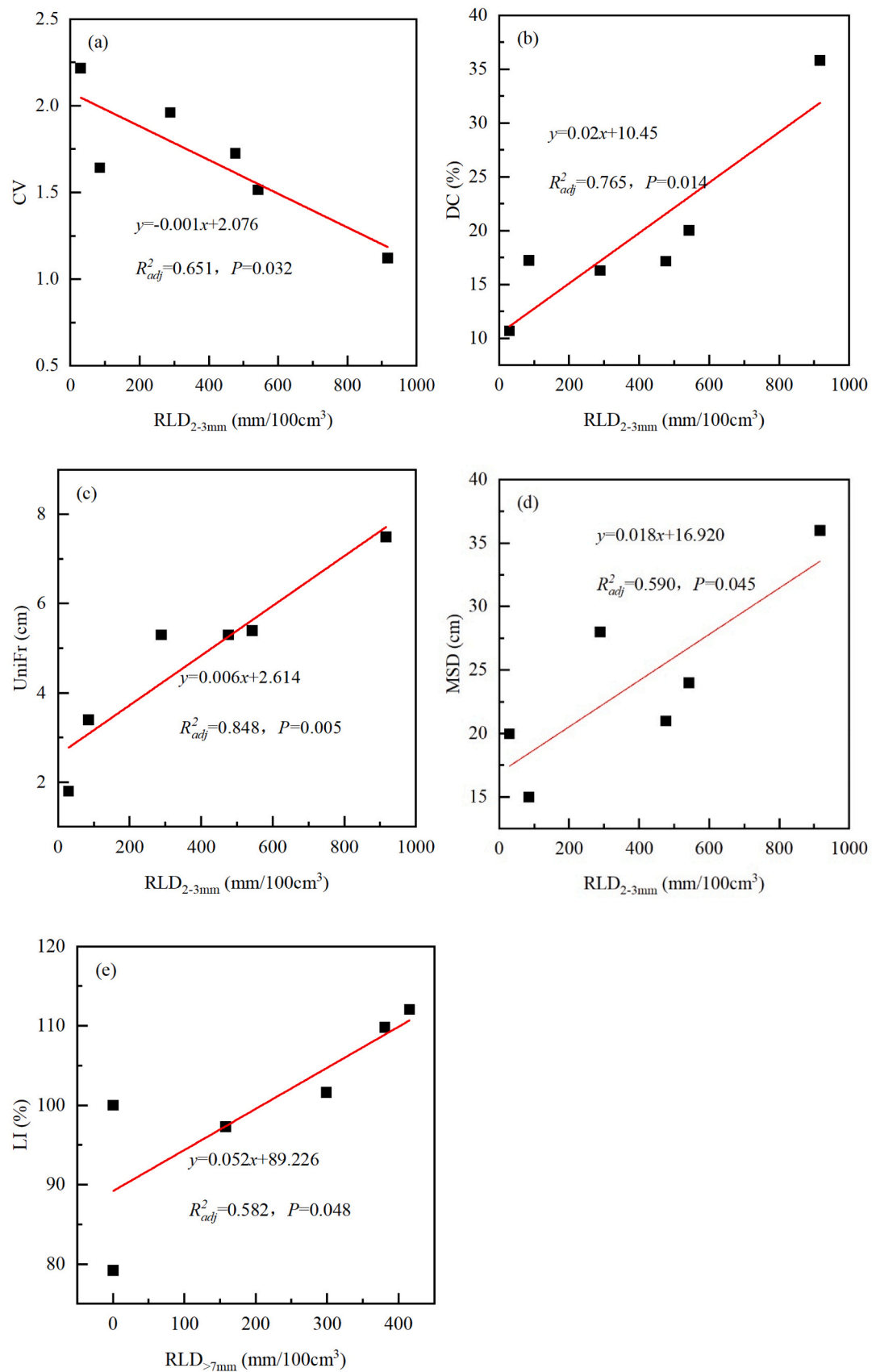


Fig. 9. Relationship between root length distribution and preferential flow parameters fitted for different diameter classes. (a) coefficient of variation (CV), (b) dye coverage (DC), (c) uniform infiltration depth (UniFr), (d) maximum stained depth (MSD), and (e) length index (LI). The regression analysis is only presented if it is significant at $P < 0.05$.

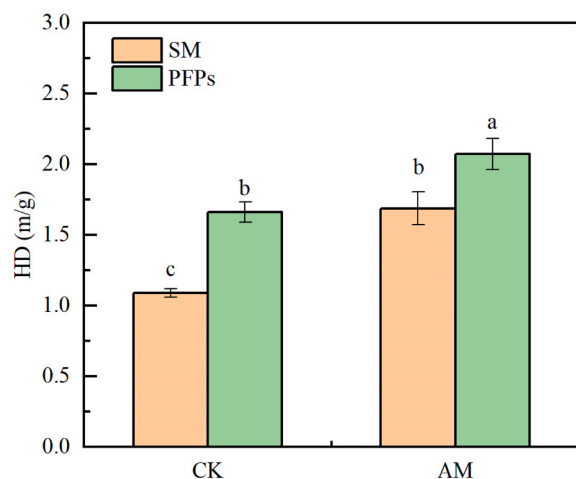


Fig. 10. HD, hyphal density. SM is the soil matrix, and PFPs are the preferential flow pathways. AM: *Amorpha fruticosa* inoculated with the AM fungus *Funneliformis mosseae*; CK: *Amorpha fruticosa* without AM fungal inoculation (Control). Different lowercase letters indicate significant differences between groups at $P < 0.05$.

exhibited a substantial increase in height, about 1.4 times that of the controls. Additionally, the RLD in the AM treatment was 1.32 times that of the control in the V3 profile, indicating that the AM fungus increased RLD at all diameters. While fine roots are conventionally attributed to nutrient acquisition and mycorrhizal mycelium is deemed capable of mineral absorption, it is noteworthy that root systems adjusted themselves to a higher proportion of large-diameter roots (Pregitzer et al., 2002). Fine roots are defined in most studies as root systems with diameter ≤ 2 mm (Chen et al., 2004; Finér et al., 2011). Notably, AM fungus inoculation did not significantly affect fine roots, but it substantially enhanced root systems with diameters > 2 mm, particularly in the 2–3 mm category, resulting in enhancements of up to 196 % (Fig. 8). The capacity of the root system to overcome soil strength is determined by the maximum expansion generated within the root elongation zone, root tip morphology, and frictional characteristics (Whalley et al., 2008). AM fungus inoculation undoubtedly enhanced the ability of the root systems to overcome soil strength. In particular, AM fungus inoculation enhanced the RLD of the root system with a diameter of 2–3 mm by nearly two times, which may be attributed to the interaction between plant roots and mycorrhizal network, potentially a response to the compacted soil conditions in the opencast coal mine dump. This augmentation in larger roots may be due to the mycelium's ability to replace the fine root system, thereby increasing the proportion of roots with diameters ≥ 2 mm. Furthermore, roots with diameters of 2–3 mm have lower nutrient and water demands compared to larger roots, enabling them to better adapt to the soil strength of the opencast coal mine waste area. This adaptation led to the preferential development of root systems with diameters of 2–3 mm following inoculation of *A. fruticosa* with *F. mosseae*.

4.3. Effects of roots and mycelia on preferential flow characteristics

Previous studies on preferential flow have often emphasized the importance of fine root systems with a diameter of < 2 mm in shaping preferential flow pathways (Raizada et al., 2013; Zhang et al., 2015b). Fine roots tend to have a short lifespan, typically ranging from a few months to a few years. As fine roots decay they create pores that enable the formation of continuous preferential flow pathways. The decomposition of these fine roots releases substantial nutrients including organic matter into the soil. The organic matter promotes microbial activity within the soil and contributes to the formation of soil aggregates. These biological, chemical, and physical changes in the soil

further enhance soil pore structure and improve its overall infiltration (Backnäs et al., 2012; Ghestem et al., 2011). Moreover, new plant roots tend to grow alongside the decaying root systems, and these newly developed roots possess greater penetration capacities than their predecessors. This iterative growth process further expands pore diameter and length (Yavitt et al., 2011). Here, we undertook a more detailed analysis of plant root diameters and observed that the RLD values of all levels of root systems in PFPs were higher than those in the SM zones. Moreover, RLDs of fine root systems with diameters < 4 mm exhibited significant positive correlations with CV, DC, and UniFr, aligning with previous findings (Raizada et al., 2013; Zhang et al., 2015a). However, it must be noted that the effects of these fine root systems were primarily concentrated in the surface soil. Conversely, root systems with diameters 2–3 mm were significantly positively correlated with MSD, and root systems with diameters > 7 mm were significantly positively correlated with LI. This suggests that root systems with diameters > 2 mm play a pivotal role in driving deep preferential flow in the opencast coal mine dump. Effective root growth requires a continuous network of appropriately sized soil pores without which root tips must exert substantial force to displace soil particles and create channels (Smith, 2007). Given the compact nature of opencast coal mine waste soils, root systems with diameters < 2 mm lack the ability to penetrate the deeper soils with higher root tip strength. Consequently, the interface between the coarse root system and the soil forms a continuous network that facilitates plant-soil-water interactions and the formation of continuous preferential flow pathways. Interestingly, roots with diameters of 3–7 mm did not have substantial effects on deep preferential flow, likely due to the short reclamation period (four years) of the dump and the smaller proportion of roots within this size range, which did not demonstrate statistically significant correlations with preferential flow indicators.

The presence of root systems and mycelia considerably influences preferential flow development in the opencast coal mine dump. Here, we observed that this influence results from the dual effects of enhanced plant root growth facilitated by the mycelial network and the alteration of soil pore structures by the mycelium. This agrees with Daynes et al. (2013) who demonstrated that mycorrhizal hyphae possess the capacity to further alter the distribution of pore sizes in the soil. The role of plant roots in stabilizing large aggregates is rather limited. The majority of the increase in water-stable aggregates (> 2 mm) can be attributed to the binding of smaller soil particles by AM fungal hyphae rather than roots (Tisdall and Oades, 1979). The effect of the root system on agglomerates often results from a combined effect of roots and mycelium. Studies show that when fungal growth is constrained relative to plant growth, large agglomerates decline by 40 %. Importantly, other agglomeration mechanisms cannot compensate for this reduction in aggregate formation caused by mycelium loss (Beare et al., 1997; Wilson et al., 2009). Remarkably, inoculation with *F. mosseae* substantially enhanced the HD by 25 % in the PFPs and by 55 % in the SM zone, suggesting that AM fungus inoculation significantly extended the distribution range and increased the density of the mycelial network. However, it is noteworthy that direct evidence supporting the formation of a mycelium-soil interface, similar to the root-soil interface, remains lacking.

5. Conclusions

The development of preferential flow in the opencast coal mine dump was driven mainly by *A. fruticosa*. Proximity to *A. fruticosa* was correlated with a higher degree of preferential flow development. Additionally, AM fungus inoculation effectively increased various preferential flow indicators. The spatial distribution of the root systems in the opencast coal mine dump was influenced mainly by *A. fruticosa*. The closer to *A. fruticosa* individuals the more abundant the plant roots and the wider the root distribution. The critical condition for preferential flow was the simultaneous presence of roots with diameter > 2 mm at both 0–10 cm and 10–20 cm soil depths. The content of roots in the stained area of each pathway was significantly higher than that in the

unstained area. AM fungus inoculation increased the content of roots in each pathway, resulting in an increase ranging from 34 % to 196 %. The influence of roots extended across different diameters, such as the staining area ratio and the coefficient of variation of the staining area ratio. Substrate infiltration was affected mainly by roots with diameters < 4 mm, and the deep preferential flow was affected by roots with diameters > 2 mm.

In summary, shrubs like *A. fruticose* are particularly suitable for soil improvement in opencast coal mine dumps. AM fungal inoculation promotes root growth in *A. fruticose*, facilitating the formation of soil preferential flow pathways, thereby enhancing rainwater retaining around roots and promoting plant growth in arid environments.

CRediT authorship contribution statement

Kejing Yin: Software, Investigation. **Xinpeng Du:** Writing – original draft, Investigation, Formal analysis, Data curation. **Yinli Bi:** Funding acquisition, Conceptualization. **Mingchao Li:** Investigation, Data curation. **Lexuan Tian:** Software, Investigation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

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